NURSING DEPARTMENT

Policy Name : Lumbar Puncture, Assisting with

Policy Number : LPA- NICU 2/3 (8)

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1. **Purpose**
    a. Obtain CSF for examination purposes such as protein, glucose, analysis and culture
    b. Measuring cerebrospinal pressure and assisting detection of observation of CSF circulation.
2. **Scope**
    a. **First Step:**
        i. Physician explains the procedure and obtains consent.
    b. **Last Step:**
        i. Appropriate sample sent to the laboratory.
3. **Definition**
    a. Lumbar Puncture is a diagnostic and at times therapeutic procedure that is performed in order to collect a sample of cerebrospinal fluid (CSF) for biochemical, microbiological, and cytological analysis, or to relieve increased intracranial pressure.
    b. Cerebrospinal Fluid is a clear, colorless, bodily fluid that occupies the subarachnoid space and the ventricular system around and inside the brain and spinal cord.
4. **Abbreviation**:
    a. LP – Lumbar Puncture
    b. CSF – Cerebrospinal Fluid
5. **Policy Statements**
    a. Procedure must be performed by a physician with written order and must be assisted by a registered staff nurse.
    b. NICU admission consent must be signed by the parents upon admission.
    c. Procedure requires sterile technique and careful patient positionin8.
    d. Lumbar puncture should not be done to patient with lumbar deformity and infection at the site of puncture.
    e. It should be performed cautiously in patient with increased intracranial pressure.
    f. Prepare sedation drugs as preferred by the physician.
    g. Emergency kit / medications must be available.
6. **Procedure and Rationale:**
    a. Material & Equipment
        i. Over bed table
        ii. Sterile gloves for the physician and the nurse.
        iii. Sterile gowns and eye sheet
        iv. Povidone-iodine solution
        v. Sterile gauze pads
        vi. Alcohol swabs
        vii. Local anesthesia
        viii. Syringes

ix. Supplemental oxygen
x. Adhesive bandage
xi. Three sterile collection tubes with stoppers
xii. Laboratory request form
xiii. Labels
xiv. Emergency Medications

S.No.	Procedure	Rationale
Preparatory Phase		
1	Identify the patient by using two identifiers (Name and medical record number - first verification)	Correct identification of patient as per IPSG goal 1.
2	Explain the procedure to the patient or parents of infant. Be sure a consent form has been signed.	To alleviate anxiety and gain cooperation.
3	Some patient may need sedation before the procedure. And if patient receive sedation observe for any respiratory problems.	Small babies tend to be agitated without sedation.
4	Keep the oxygen beside the patient and make sure that Emergency kits such as intubation tray and drugs are available.	For emergency purposes.
5	Make sure that the patient is properly connected to monitor.	
6	Before procedure, the patient should empty his/her bladder and bowel if possible.	To enhance comfort.
7	Hygienic wash hand.	To prevent cross infection.
8	Open the equipment tray on a table and be careful not to contaminate sterile field when you open the wrapper.	To prevent cross contamination.
9	`Provide adequate lighting at the puncture site and adjust the height of the patient's bed.	To provide comfort to the surgeon who will perform for the procedure.
10	Position the patient on side with pillow under head and a pillow between legs, patient should lie on a firm surface. For small babies, both legs must be grabbed and flexed to abdomen and the head must be flexed forward by holding the shoulder making sure not to block the airway.	The spine is maintained in a horizontal position. The pillow between the legs prevents the upper leg from rolling forward.
11	For big children instruct the patient to arch the lumbar segment of back and draw knees up to abdomen, chin to chest, clasping knees with hands.	This posture offers maximal widening of the inter-spinous spaces and affords easier entry into the subarachnoid space.
12	Assist the patient in this maintaining position by supporting behind the knees and neck	To prevent sudden movement which may produce traumatic tap thus impede the correct diagnosis.

	throughout the examination.	
13	Don sterile gloves both doctor and nurses.	To maintain procedure sterile.
Performance Phase		
14	Clean the puncture site with sterile gauze pads soaked in povidone-iodine solution wiping in circular motion away from the puncture site.	To prevent spread of infection.
15	Drape the area with sterile eye sheet and hand towel.	To maintain the sterile field along the procedure.
16	Before spinal needle is anesthetic is administered puncture.	To decrease patient discomfort.
17	When the doctor inserts the spinal needle into subarachnoid space between the third and fourth vertebral, hold the patient firmly in position.	L3-L4 interspaces are below the level of the spinal cord.
18	Collect about 1ml of CSF in each of the sterile specimen tubes by allowing the fluid to drip into the tubes, for laboratory analysis, spinal fluid should be clear and colorless	Bloody spinal fluid may indicate subarachnoid hemorrhage or a traumatic tap.
Follow-up Phase		
19	After procedure, the patient is instructed to remain flat for about 2 hours.	To reduces the pressure on tissue surfaces along the needle track to prevent CSF leakage.
20	Ensure adequate hydration with oral or parenteral fluid.	This facilitates replacement of CSF and prevent spinal headache
21	Monitor the spinal headache and observe for CSF leak.	CSF from lumbar puncture are characterized by intractable headache.
22	Documents the following: a. Initiation and completion b. Time of the procedure c. Number of specimen tubes Collected d. The time transport to laboratory e. Color, consistency and other characteristics of the collected sample.	Proper documentation will serve as an evidenced that the intervention has been done.
23	Results should be relayed the physician as soon as possible	To provide appropriate i needed.

7. **Responsibilities**
   a. **Physician** is responsible for performing and collecting CSF sample under aseptic technique.

b. **Head Nurse** is responsible to make sure that all needed equipment are available in the unit and make sure that her staff is competent in following strict septic procedure during the procedure.
c. **Charge Nurse** is responsible to check physician's order if carried.
d. **Staff Nurses** are responsible to carry out physician's orders. Assisting in LP procedure and ensure the safety of patient before and after the procedure.
e. **Laboratory Technician** is responsible to run and test all the samples brought to them and releases the results correctly.

8. **Documentation**
   a. NICU Flowchart
   b. Physician Progress notes
   c. Physician Order sheet
   d. NICU Kardex
9. **Measurements**
   a. 100% nurses can assist the doctor in doing the procedure.
   b. 100% of all requests completed and signed.
10. **References**
    a. Ministry of health Policy and Procedure, KSA
    b. Textbook of Medical Surgical Nursing Brunner & Suddarth's, 10th edition.
11. **Cross References**
    N/A
12. **Attachments**
    N/A
13. **Distribution List**
    a. Nursing Administration
    b. Quality Department
    c. Office of the Executive Director General
14. **Review Summary**
    a. Restatement of scope.
    b. Revision of some procedure steps.